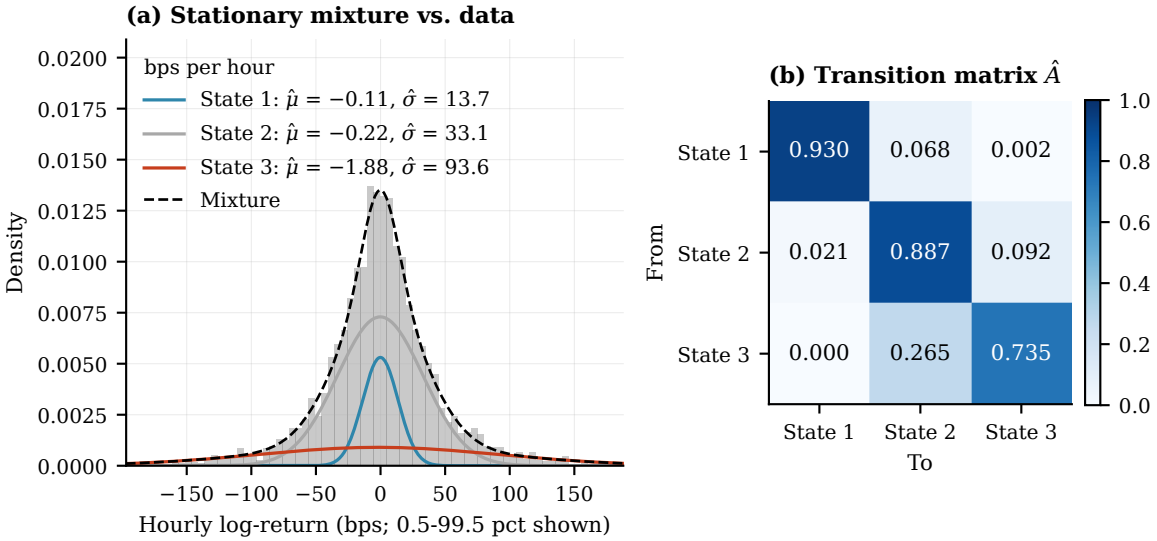
(a) Stationary mixture vs. data
bps per hour
State 1: $\hat{\mu} = -0.11$, $\hat{\sigma} = 13.7$
State 2: $\hat{\mu} = -0.22$, $\hat{\sigma} = 33.1$
State 3: $\hat{\mu} = -1.88$, $\hat{\sigma} = 93.6$
Mixture
Density
0.0200
0.0175
0.0150
0.0125
0.0100
0.0075
0.0050
0.0025
0.0000
−150 −100 −50 0 50 100 150
Hourly log-return (bps; 0.5-99.5 pct shown)
(b) Transition matrix $\hat{A}$
From
State 1
State 2
State 3
0.930
0.068
0.002
0.021
0.887
0.092
0.000
0.265
0.735
State 1
State 2
State 3
To
1.0
0.8
0.6
0.4
0.2
0.0